MILITARY SPECIFICATION

ARCTICCUSHIONS - CLOTHING, CHU-10/P AND CHU-11/P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers arctic cushions - clothing, which consist of extreme cold weather survival clothing that has been packaged in a cover to a specified seat comfort cushion configuration. The arttic cushions - clothing are designated CHU-10/P and CHU-11/P (see 3.5 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-375	Plastic Film, Flexible, Vinyl Chloride
L-P-383	Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated
V-T-295	Thread, Nylon
KK-L-162	Leather, Cattlehide, Chrome Tanned for Gloves, Garments and Equipage
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-T-60	Tape: Packaging, Waterproof

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-1-6903	Ink, Marking (for Parachutes and Other Textile Items)
MIL-C-20696	Cloth, Coated, Nylon, Waterproof
MIL-F-21840	Fasterner Tapes, Hook and Pile, Synthetic
MIL-O-27525	Overcoat, Survival, Arctic SRU-6/P
MIL-M-27529	Mittens, Extreme Cold Weather Survival, SRU-10/P
MIL-F-27532	Footwear Covers, Extreme Cold Weather Survival SRU-13/P
MIL-T-38328	Tape, Textile, Nylon, Aromatic, Nonmelting, Reinforcing
MIL-T-43636	Thread, Nylon, Non-Melting

STANDARDS

Federal

FED-STD-751	Stitches, Seams, and Stitchings

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging of and Packing of
MIL-STD-831	Test Reports, Preparation of
MS20230	Grommets, Metallic, Plain and Spur, with Washers, Type I and Type III
MS24665	Pin, Cotter (Split)
MS27983	Fasteners, Snap, Style 4 (Three Way Locking Type.)

DRAWINGS

Air Force

75116	Button, Tufting
75117	Plate, Base
75118	Insert, Arctic
75119	Cover, Arctic Cushion - Clothing CHU-10/P & CHU-11/P
75120	Arctic Cushion - Clothing CHU-10/P & CHU-11/P

(Copies of specifications, standards, drawings, and publication required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article inspection. This specification makes provisions for first article inspection (see 4.4 and 6.2).

3.2 Major components. Each CHU-10/P and each CHU-11/P arctic cushion - clothing shall consist of the following major components:

(a) Insert

(b) Cover.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Basic fabric for cover. The basic frabic for the cover shall conform to type III, class 5 of MIL-C-20696 except coated with chloroprene containing fire retardant. The basic frabic for the cover shall be sage green, USAF color shade No. 1535 (see 6.3).

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3.4.2 Leather. The reinforcements for the cover and the tufting buttons for the insert shall be made of leather that conforms to type I, cLASS 1 of KK-L-162 and shall be 1/16 inch thick. The color of the leather shall be black.

3.4.3 Thread. The thread for sewing the cover shall conform to type I, size F of MIL-T-43636. The color of the thread shall be sage green, USAF color shade No. 1535 (see 6.3).

3.4.4 Fastern tape. The hook and the pile fastener tape for the cover shall be sage green, US Navy color shade No. 3411 (see 6.3) and shall be the width specified on Drawing 75119. The hook fastener tape shall conform to type I, class 2 of MIL-F-21840. The pile fastener tape shall conform to class 2 of MIL-F-2184O.

3.4.5 Nylon tape. The carrying handles for the cover shall be made of nylon tape that conforms to type 11 of MIL-T-38328. The color of the nylon tape shall be yellow, USAF color shade No. 1365 (see 6.3).

3.4.6 Snap fasteners. The snap fasteners for the cover shall conform to MS27983-1 and MS27983-2.

3.4.7 Slide fasterners. The slide fasteners for the cover of the CHU-10/P arctic cushion - clothing shall be as specified for the -10 configuration on Drawing 75119. The slide fastener for the cover of the CHU-11/P arctic cushion - clothing shall be as specified for the -30 configuration on Drawing 75119 (Note: Although only one-half of the separating slide fastener is sewn to the cover, each cover shall be provided with both the left-hand half and the right-hand half of the slide fastener.)

3.4.8 Cord loops for tufting. The cord loops for tufting shall be nylon thread that conforms to type II, class 1, size No. 6 of V-T-295. The color of the nylon thread shall be sage green, USAF color shade No. 1535 (see 6.3).

3.4.9 Grommets. The grommets for the base plate shall conform to MS20230BP20.

3.4.10 Laminated plastic material. The base plate shall be made of laminated plastic material that conforms to type I, color natural, 0.06 inch thick of L-P-383.

3.4.11 Release pin. The release pin for the insert shall conform to MS24665-308.

3.4.12 Plastic tape. The plastic tape for the insert shall conform to type III, class 2 of PPP-T-60.

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3.5 Design and construction. The CHU-10/P arctic cushion - clothing shall be designed and constructed in accordance with the -10 configuration on Drawings 75116 through 75120 and as specified herein. The CHU-11/P arctic cushion - clothing shall be designed and constructed in accordance with the -30 configuration on Drawings 75116 through 75120 and as specified herein. In the event of conflict between the requirements of these drawings and the text of this specification the text of this specification shall govern.

3.5.1 Assembly of major components. The major components of the arctic cushion - clothing shall be assembled in accordance with rawing 75120 as follows: The bag shall be cut away from the insert specified in 3.7.2. The excess plastic shall be trimmed. The insert shall then be placed inside the cover specified in 3.7.1 and shall be restrained in the final configuration by the cover. The flaps of the cover shall be closed by means of the fastener tape.

3.6 Performance

3.6.1 Disassembly. The extreme cold weather survival clothing shall be tufted and placed in the cover so that one person who is wearing flight gloves can disassemble the arctic cushion - clothing and restore the extreme cold weather survival clothing to donnable condition within 30 seconds,

3.6.2 Impact resistance. After the arctic cushion - clothing has been dropped three times from a height of 10 feet onto a smooth concrete floor, the arctic cushion - clothing shall not show any evidence of closure deterioration or shape deformation.

3.7 Details of components

3.7.1 Cover. The cover shall conform to the -10 configuration on Drawing 73119 for the CHU-10/P arctic cushion - clothing or to the -30 configuration on Drawing 75119 for the CHU-11/P arctic cushion - clothing as applicable. The cover shall have a two-flap closure for easy access to the packaged clothing, two carrying handles, and provisions (slide fastener and snap fasteners) for attachment to a survival kit container,

3.7.1.1 Sewing Operations. The stitching used for the cover shall conform to stitch type 301 of FED-STD-751, shall be 7, to 11 stitches per inch, and shall be a minimum of 1/8 inch from the edge. The ends of all stitchings shall be backstitched not less than 1/2 inch.

3.7.2 Insert. The insert shall consist of extreme cold weather survival clothing that has been vacuum packaged and attached to a base plate by tufting buttons as follows:

(a) The extreme cold weather survival clothing shall be tumble dried in a clothes dryer at an elevated temperature (plus 150 degrees Fahrenheit minimum, for a minimum of 30 minutes) to remove any moisture that may be present,

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(b) Immediately after drying. the extreme cold weather survival clothing shall be folded and arranged as specified on Drawing 75118 to permit reduction to the size and shape required. The clothing components-shall be anchored or connected to prevent accidental loss.

(c) The extreme cold weather survival clothing shall be placed in a bag. The bag shall be made of plastic that is approximately 12 mil thick and that is equivalent to the plastic film specified in L-P-375. While the extreme cold weather survival clothing and the bag are being shaped to the desired size, the air in the bag shall be evacuated. The bag shall be heat sealed.

(d) The extreme cold weather survival clothing shall be attached to the base plate by tufting buttos, as shown on Drawing 75118, so that it will maintain the required shape and thickness. Because the vacuum in the bag will be lost during the insertion of the tufting buttons, the bag containing the extreme cold weather survival clothing shall be mounted in a frame to hold the required shape during the tufting.

3.7.2.1 Extreme cold weather survival clothing. The extreme cold weather survival clothing shall be either sage green or Indian orange as specified by the procuring activity (see 3.9 and 6.2 and shall be as follows:

(a) One SRU-6/P overcoat that conforms to class 1 of 2 of MIL-O-27525

(b) One pair of SRU-10/P mittens that conform to class 1 or 2 of MIL-M-27529

(c) One pair of SRU-13/P footwear covers that conform to class 1 or 2 or MIL-F-27532.

3.7.2.2 Tufting buttons. The tufting buttons shall conform to Drawing 75116.

3.7.2.3 Base plate. The base plate shall conform to the -10 configuration on Drawing 75117 for the CHU-10/P arctic cushion - clothing or the -30 configuration on Drawing 75117 for the CHU-11/P arctic cushion - clothing as applicable.

3.8 Dimensions. The CKU-10/P arctic cushion - clothing shall conform to the envelope dimensions specified for the -10 configuration on Drawing 75119. The CHU-11/P arctic cushion - clothing shall conform to the envelope dimensions specified for the -30 configuration on Drawing 75119.

3.9 Government furnished property. When specified (see 6.2), the extreme cold weather survival clothing specified in 3.7.2.1 will be furnished by the Government.

3.10 Operational instructions and arrows. The operational instructions and the arrows shown on Drawing 75119 shall be marked in ink on the cover. The ink shall conform to type IV of MIL-I-6903 and shall be yellow, USAF color shade No. 1365 (see 6.3).

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3.11 Identification of product. Except for the extreme cold weather survival clothing, the arctic cushion - clothing shall be marked for identification in accordance with MIL-STD-130.

3.12 Workmanship. The CHU-10/P and CHU-11/P arctic cushions - clothing shall be constructed in a thoroughly workmanlike manner, The accuracy of dimensions, proper marking of assemblies, and overall comfort of each arctic cushion - clothing shall be given particular attention,

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein, Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examinations and the tests required by this specification are classified as follows:

(a) First article inspection

(b) Quality conformance inspection.

4.3 Atmospheric conditions, Unless otherwise specified, all examinations and tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77 (plus or minus 18) degrees Fahrenheit, and at a relative humidity of 80 percent or less. Where examinations and tests are made with an atmospheric pressure or temperature substantially different from these values, proper allowance shall be made for the change in instrument reading,

4.4 First article inspection. The first article inspection shall consist of all of the examinations and tests described under 4.6.

4.4.1 First article samples. Before production is commenced, the contractor shall manufacture the first article samples specified in 4.4,1.1 and 4.4,1.2. The-first article samples shall be manufactured in the facilities to be used for the production items. The first article samples shall be representative of the materials, components, design, construction, dimensions, and workmanship to be used in the production items. Approval of the first article samples will authorize commencement of production but shall not preclude the requirement for the quality conformance inspection,

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4.4.1.1 Tested samples. The contractor shall subject one arctic cushion - clothing of each type on the contract or order, which has been manufactured as specified in 4.4.1, to the first article inspection (see 6.2).

4.4.1.2 Untested samples. The contractor shall submit one untested arctic cushion - clothing of each type on the contract or order, which has been manufactured as specified in 4.4.1, to the procuring activity (see 6.2).

4.4.2 Test report. When required (see 6,2), the contractor shall submit a first article test report that has been prepared in accordance with MIL-STD-831.

4.5 Quality conforrmance inspection. Quality conformance inspection shall consist of the following as described herein:

(a) Material and component inspection

(b) End item inspection

(c) Preparation for delivery inspection.

4.5.1 Material and component inspection. Unless otherwise specified (see 6.2), materials and components to be used in the manufacture of the arctic cushions - clothing shall be inspected in accordance with the applicable provisions of the documents referenced herein. When specified by the procuring activity (see 6.2), certificates of compliance with the requirements specified herein for the applicable materials and components shall be furnished by the supplier. The certificates of compliance shall be accompanied by actual test or inspection reports or other verifiable data.

4.5.2 End item inspection. The end item inspection shall consist of sampling tests.

4.5.2.1 Sampling tests. The lot definition, formation, and size for the sampling tests shall be in accordance with MIL-STD-105. The sample unit and the unit for the expression of the lot size for the sampling tests shall be one completely assembled arctic cushion - clothing. Samples of the arctic cushion - clothing shall be selected in accordance with inspection level III of MIL-STD-105 and subjected to the examination of product described in 4.6.1 The acceptable quality level for the sampling tests shall be 2.5 defects per 100 units.

4.5.3 Preparation for delivery inspection. Inspection to determine compliance with the requirements specified herein with respect to packaging, packing, and marking shall be performed in accordance with MIL-P-116.

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4.6 Test methods

4.6.1 Examination of product, The completely assebled artic cushion - clothing shall be visually examined for the defects listed in table I.

TABLE I. Examination of Product

Examine	Defects
Arctic cushion - clothing	Any required detail of construction omitted or not as specified
	Identification marking or operational marking omitted, incorrect, illegible, or improperly applied
	Closure loosely or insecurely applied (not as specified)
	Bulging or distortion of assemblies
	Any component missing
	Any component damaged
	Any material not as specified
	Any operation omitted
	Any foreign matter or material that is not readily removable and that might cause damage to material or affect intended use
	Any dimension not as specified on drawings
	Noncompliance with requirements of 3.7.2(a).

4.6.2 Disassembly. The temperature of the artic cushion - clothing and its surrounding atmosphere during this test shall be zero degree (plus or minus 10 degrees) Fahrenheit, This test shall be performed by a test subject of average size and strength and dressed in flight clothing as specified and provided by the procuring activity (see 6.2). The test subject using only one leg and two fingers of one gloved hand (of one arm only) to apply force, shall completely remove and restore the extreme cold weather survival clothing to a donnable condition within 30 seconds from the start of this test. After this tests the extreme cold weather survival clothing shall not show any visible evidence of damage except where tufting was performed.

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4,6.3 Impact resistance. The arctic cushion - clothing shall be dropped three times from a height of 10 feet onto a smooth concrete floor such that it will strike the floor in three different random positions. After this test, the article cushion - clothing shall not show any evidence of closure deterioration or shape deformation.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A The artic cushions - clothing shall be preserved in accordance with method IC of MIL-P-116 and packaged in accordance with level A packaging of MIL-STD-794.

5.1.2 Level C. The artic cushions - clothing shall be preserved in accordance with method IC of MIL-P-116 and packaged in accordance with level C packaging of MIL-STD-794.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. The artic cushions - clothing shall be packed, in accordance with level A packing of MIL-STD-794, in boxes that conform to class weather-resistant of PPP-B-636.

5.2.2 Level C. The article cushions - clothing shall be packed to afford adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to the rules of carriers and regulation as applicable to mode of transportation and may be the contractor's commercial practice, provided the later meets the requirements for this level.

5.3 Marking. In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The CHU-10/P and the CHU-11/P arctic cushions - clothing covered by this specification are intended to be used as comfort cushions on the top of survival kit containers and to provide clothing to be worn in extreme cold weather. The CHU-10/P arctic cushion - clothing is intended to be used with the survival kit container in the F-101 aircraft. The CHU-11/P artic cushion - clothing is intended to be used with the survival kit container in the F-102 or the F-106 aircraft.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Either CHU-10/P or CHU-11/P arctic cushion - clothing (see 1.1)

(c) Requirements for first article inspection including test report and whether or not" tested first article samples should be retufted, re-assembled, and submitted (see 3.1 and 4.4 through 4,4,2)

(d) Either class 1 (sage green) or class 2 (Indian orange) for the color of the extreme cold weather survival clothing (see 3.7.2,1)

(e) Requirements for Government furnished extreme cold weather survival clothing (see 3.9)

(f) Requirements for material and component inspection and certificates of compliance (see 4.5.1)

(g) Requirements for Government furnished flight clothing for test subject (see 4.6.2)

(h) Selection of applicable levels of preservation and packaging and packing (see 5.1 and 5.2).

6.3 Color samples. Color samples can be obtained from the procuring activity or as directed by the contracting officer.

Custodian:
Air Force - 11

Reviewer:
Air Force - 71

Preparing activity:
Air Force - 11

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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